Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Wagner Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
20623 E Lone Tree RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1992

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X110-XX04-XXXX

B. OPERATORS

Wagner, Dustin

Operator name: Wagner, Dustin

Telephone no.: (209) 838-7547 Landline, (209) 652-5192 Cellular

Mailing Address Number and Street	City	State	Zip Code
22176 Skiff RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Brasil, Victor

Legal owner name: Brasil, Victor

Telephone no.: Landline, (209) 059-5157 Cellular

Mailing Address Number and Street	City	State	Zip Code
21728 Lone Tree RD	Escalon	CA	95320

Brasil & Sons Dairy

Legal owner name: Brasil & Sons Dairy

Telephone no.: Landline, (209) 595-1577 Cellular

Mailing Address Number and Street	City	State	Zip Code
21768 E Lone Tree RD	Escalon	CA	95320

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,100	0	0	0	0	0
Maximum number	1,100	0	0	0	0	0
Average number	1,100	0	0	0	0	0
Avg live weight (lbs)	1,400	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 28,736.19 *tons per reporting period*

Total nitrogen from manure: 375,669.73 *lbs per reporting period*

After ammonia losses (30% loss applied): 262,968.81 *lbs per reporting period*

Total phosphorus from manure: 63,697.06 *lbs per reporting period*

Total potassium from manure: 213,550.95 *lbs per reporting period*

Total salt from manure: 517,935.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 16,566,000 *gallons*

Total nitrogen generated: 70,004.79 *lbs*

Total phosphorus generated: 5,853.53 *lbs*

Total potassium generated: 108,280.56 *lbs*

Total salt generated: 577,757.73 *lbs*

	16,566,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	16,566,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal SSJID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/20/2023	Solid commercial fertilizer *4-0-0*	8.50 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	679.32	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	679.32	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	Corral solids	1,070.00 *ton*	Dry-weight	57.0		30,400.00	15,100.00	49,100.00		25.80

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	27,974.08	13,895.02	45,181.82	237,411.60
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	27,974.08	13,895.02	45,181.82	237,411.60

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
D1-1	61	61	2	process wastewater	X205-X110-XX04-XXXX
D1-2	63	63	2	process wastewater	X205-X110-XX04-XXXX
D1-3	20	20	2	process wastewater	X205-X110-XX04-XXXX
D1-4	7	7	2	process wastewater	X205-X110-XX04-XXXX
Manteca Xavier	85	85	2	manure	X177-X500-XX30-XXXX
Totals for areas that were used for application	236	236	10		
Totals for areas that were not used for application					
Land application area totals	236	236	10		

B. CROPS AND HARVESTS

D1-1

Field name: D1-1

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	890.60 *ton*	As-is		63.8	6,880.00	910.00	9,670.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.60	200.90	26.57	282.36	1,067.61

D1-1

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	1,652.80 *ton*	As-is		69.0	4,000.00	590.00	4,760.00		5.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.10	216.76	31.97	257.95	967.62

D1-2

Field name: D1-2

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 63 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	913.50 *ton*	As-is		63.4	4,500.00	970.00	7,310.00		9.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.50	130.50	28.13	211.99	975.43

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/30/2023	1,781.40 *ton*	As-is		69.6	3,530.00	560.00	3,220.00		3.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.28	199.63	31.67	182.10	675.64

D1-3

Field name: D1-3

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	248.00 *ton*	As-is		55.6	4,480.00	1,040.00	5,930.00		8.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.40	111.10	25.79	147.06	941.46

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	524.00 *ton*	As-is		63.4	4,680.00	480.00	5,520.00		5.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.20	245.23	25.15	289.25	1,127.69

D1-4

Field name: D1-4

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	86.80 *ton*	As-is		55.6	4,480.00	1,040.00	5,930.00		8.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.40	111.10	25.79	147.06	941.46

D1-4

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	183.40 *ton*	As-is		63.4	4,680.00	480.00	5,520.00		5.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.20	245.23	25.15	289.25	1,127.69

Manteca Xavier

Field name: Manteca Xavier

11/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 85 Plant date: 11/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	1,224.00 *ton*	As-is		63.4	4,140.00	890.00	5,490.00		8.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.40	119.23	25.63	158.11	929.70

06/20/2023: Corn, grain

Crop: Corn, grain Acres planted: 85 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	2,244.00 *ton*	As-is		66.4	3,230.00	620.00	3,510.00		5.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.40	170.54	32.74	185.33	938.49

NUTRIENT BUDGET

A. LAND APPLICATIONS

D1-1 - 11/20/2022: Oats, silage-soft dough

Field name: D1-1

Crop: Oats, silage-soft dough — Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.96	20.95	134.31	1,137.66	1,650,000.00 *gal*
Application event totals		111.96	20.95	134.31	1,137.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.29	5.86	183.79	876.36	1,518,000.00 *gal*
Application event totals		105.29	5.86	183.79	876.36	

D1-1 - 06/15/2023: Corn, silage

Field name: D1-1

Crop: Corn, silage — Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	146.49	8.15	255.70	1,219.28	2,112,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	39.61	4,992,000.00 *gal*
Application event totals		146.49	8.15	255.70	1,258.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	59.41	7,488,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.41	

D1-1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	58.18	7,332,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	55.70	7,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.06	9.53	186.43	926.92	1,914,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	35.90	4,524,000.00 *gal*
Application event totals		134.06	9.53	186.43	962.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	56.94	7,176,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	54.46	6,864,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	51.99	6,552,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.99	

D1-2 - 11/20/2022: Oats, silage-soft dough

Field name: D1-2

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.06	17.04	109.24	925.29	1,386,000.00 *gal*
Application event totals		91.06	17.04	109.24	925.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.22	4.68	147.00	700.96	1,254,000.00 *gal*
Application event totals		84.22	4.68	147.00	700.96	

D1-2 - 06/18/2023: Corn, silage

Field name: D1-2

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	141.84	7.89	247.58	1,180.57	2,112,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	38.35	4,992,000.00 *gal*
Application event totals		141.84	7.89	247.58	1,218.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	59.93	7,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.93	

D1-2 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.53	7,488,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.53	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	125.33	8.91	174.29	866.54	1,848,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	33.56	4,368,000.00 *gal*
Application event totals		125.33	8.91	174.29	900.10	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.53	7,488,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.53	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	53.93	7,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.93	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	50.34	6,552,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.34	

10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	50.34	6,552,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.34	

D1-3 - 11/23/2022: Oats, silage-soft dough

Field name: D1-3

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.62	7.77	243.72	1,162.12	660,000.00 *gal*
Application event totals		139.62	7.77	243.72	1,162.12	

D1-3 - 06/15/2023: Corn, silage

Field name: D1-3

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	153.58	8.54	268.09	1,278.34	726,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	41.53	1,716,000.00 *gal*
Application event totals		153.58	8.54	268.09	1,319.87	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.40	2,496,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.40	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	56.63	2,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.63	

D1-3 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	56.63	2,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	141.00	10.02	196.07	974.86	660,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	37.75	1,560,000.00 *gal*
Application event totals		141.00	10.02	196.07	1,012.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	54.74	2,262,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	52.85	2,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	52.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	49.08	2,028,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.08	

D1-4 - 11/23/2022: Oats, silage-soft dough

Field name: D1-4

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

D1-4 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.62	7.77	243.72	1,162.12	231,000.00 *gal*
Application event totals		139.62	7.77	243.72	1,162.12	

D1-4 - 06/15/2023: Corn, silage

Field name: D1-4

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	159.57	8.88	278.53	1,328.14	264,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	43.15	624,000.00 *gal*
Application event totals		159.57	8.88	278.53	1,371.29	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.02	897,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.02	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	59.33	858,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.33	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	53.93	780,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.93	

D1-4 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	141.00	10.02	196.07	974.86	231,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	37.75	546,000.00 *gal*
Application event totals		141.00	10.02	196.07	1,012.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	59.33	858,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	53.93	780,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	48.54	702,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.54	

Manteca Xavier - 11/21/2022: Oats, silage-soft dough

Field name: Manteca Xavier

Crop: Oats, silage-soft dough

Plant date: 11/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	151.62	70.18	213.60	1,208.88	700.00 *ton*
Application event totals		151.62	70.18	213.60	1,208.88	

Manteca Xavier - 06/20/2023: Corn, grain

Field name: Manteca Xavier

Crop: Corn, grain

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	215.30	106.94	347.74	1,827.25	700.00 *ton*
Application event totals		215.30	106.94	347.74	1,827.25	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	61.50	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.50	

06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.40	10,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.40	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	55.35	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.35	

Manteca Xavier - 06/20/2023: Corn, grain

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	59.45	10,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.45	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.40	10,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.40	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	53.30	9,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.30	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	49.20	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.20	

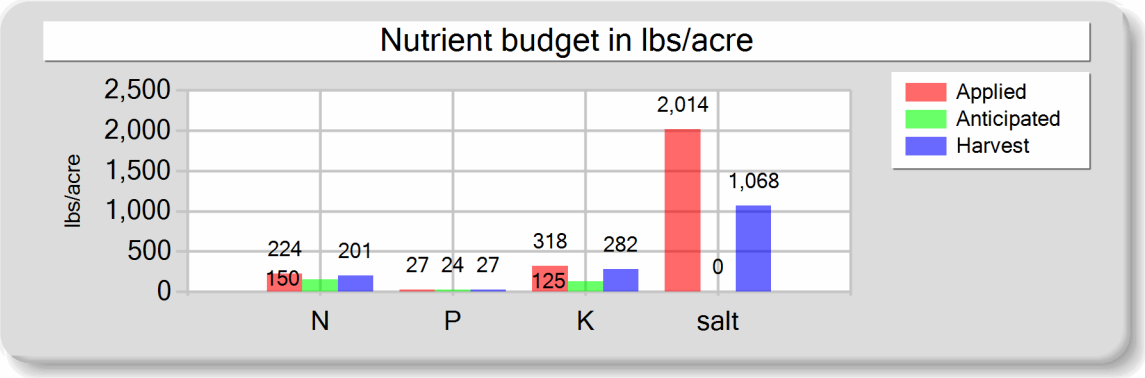
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	49.20	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.20	

B. NUTRIENT BUDGET

D1-1 - 11/20/2022: Oats, silage-soft dough

Field name: D1-1 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.25	26.80	318.09	2,014.01
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.25	26.80	318.09	2,014.01
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	200.90	26.57	282.36	1,067.61
Nutrient balance	23.35	0.23	35.73	946.40
Applied to removed ratio	1.12	1.01	1.13	1.89

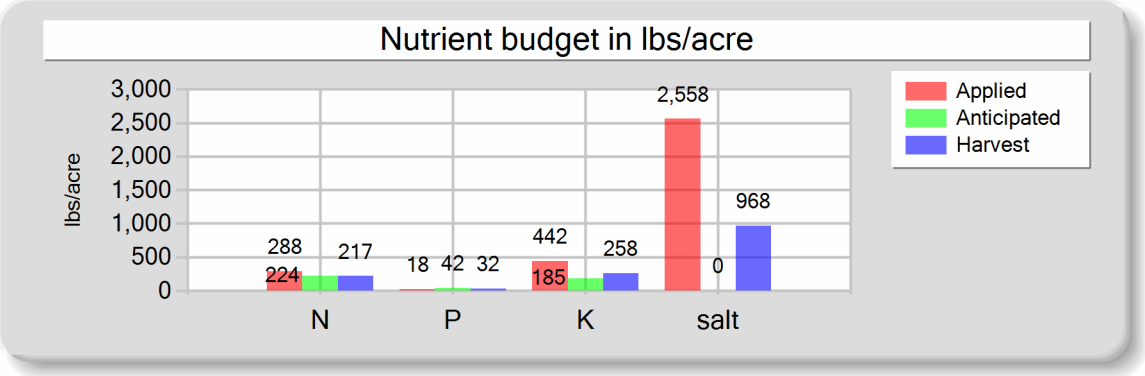
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
3,168,000.00 *gallons*
116.67 *acre-inches*
1.91 *inches/acre*

Total harvests for the crop
1 *harvests*

D1-1 - 06/15/2023: Corn, silage

Field name: D1-1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.55	17.68	442.13	2,146.20
Fresh water	0.00	0.00	0.00	412.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.55	17.68	442.13	2,558.38
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	216.76	31.97	257.95	967.62
Nutrient balance	70.79	-14.29	184.19	1,590.76
Applied to removed ratio	1.33	0.55	1.71	2.64

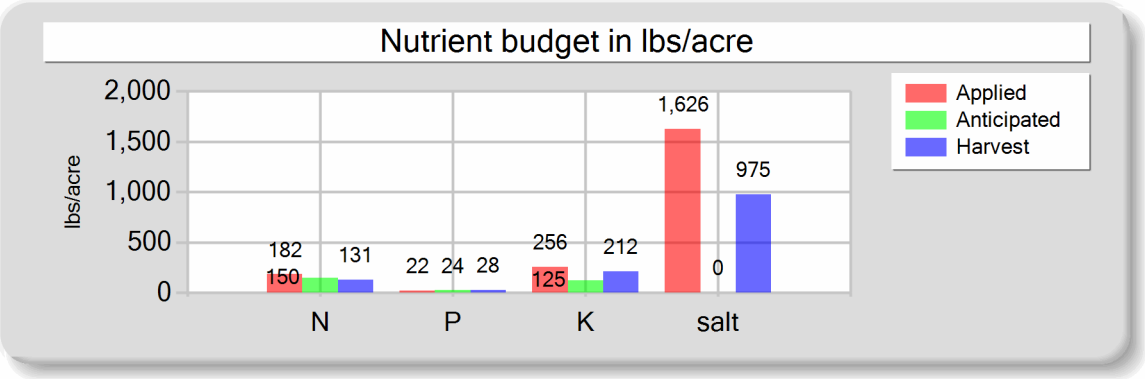
Fresh water applied	
51,948,000.00	*gallons*
1,913.07	*acre-inches*
31.36	*inches/acre*

Process wastewater applied	
4,026,000.00	*gallons*
148.26	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1-2 - 11/20/2022: Oats, silage-soft dough

Field name: D1-2 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	175.28	21.72	256.24	1,626.26
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.28	21.72	256.24	1,626.26
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	130.50	28.13	211.99	975.43
Nutrient balance	51.78	-6.41	44.25	650.83
Applied to removed ratio	1.40	0.77	1.21	1.67

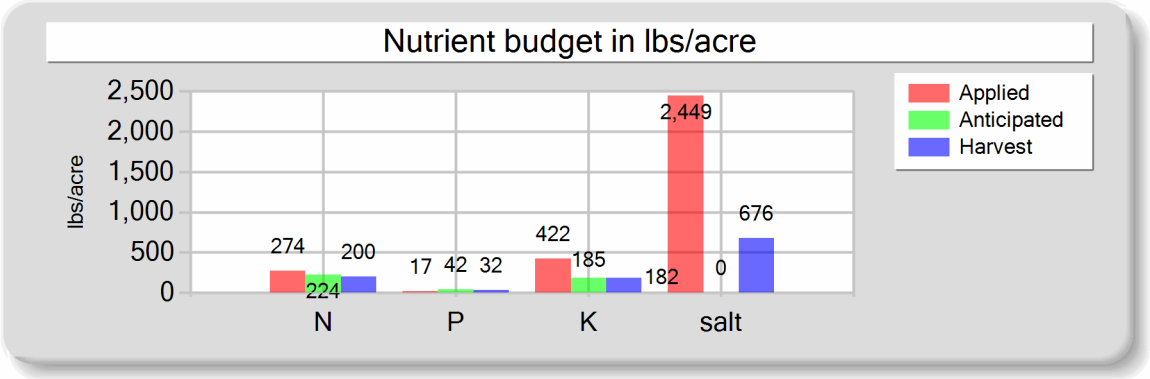
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,640,000.00	*gallons*
97.22	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1-2 - 06/18/2023: Corn, silage

Field name: D1-2 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	267.17	16.80	421.87	2,047.12
Fresh water	0.00	0.00	0.00	401.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	274.17	16.80	421.87	2,448.61
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	199.63	31.67	182.10	675.64
Nutrient balance	74.54	-14.87	239.77	1,772.97
Applied to removed ratio	1.37	0.53	2.32	3.62

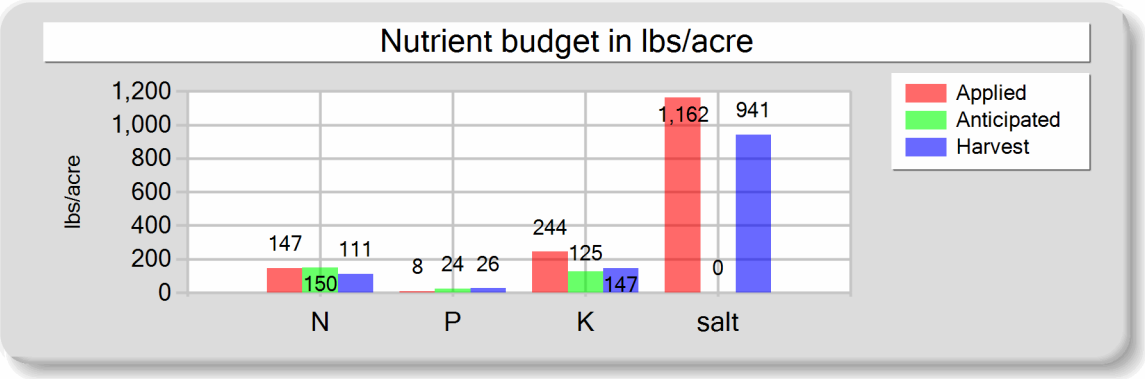
Fresh water applied	
52,260,000.00	*gallons*
1,924.56	*acre-inches*
30.55	*inches/acre*

Process wastewater applied	
3,960,000.00	*gallons*
145.83	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1-3 - 11/23/2022: Oats, silage-soft dough

Field name: D1-3 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	139.62	7.77	243.72	1,162.12
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.62	7.77	243.72	1,162.12
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	111.10	25.79	147.06	941.46
Nutrient balance	35.52	-18.03	96.65	220.67
Applied to removed ratio	1.32	0.30	1.66	1.23

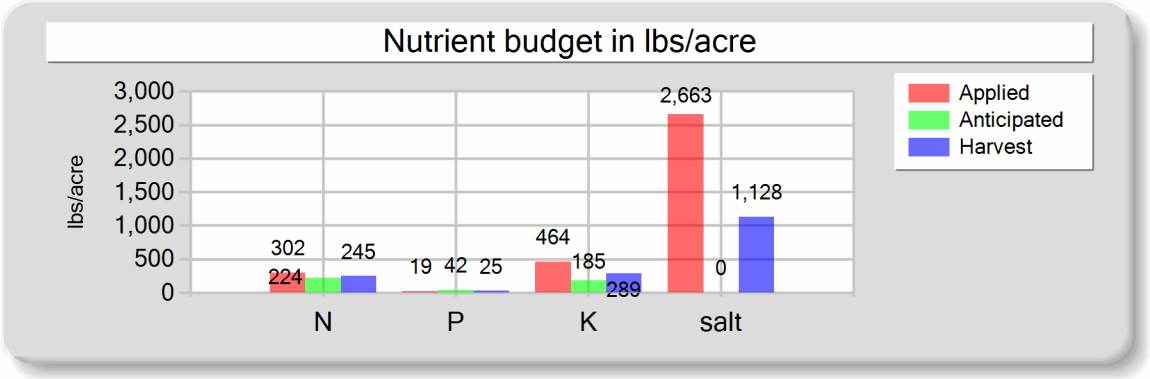
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
660,000.00	*gallons*
24.31	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1-3 - 06/15/2023: Corn, silage

Field name: D1-3 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.58	18.57	464.16	2,253.20
Fresh water	0.00	0.00	0.00	409.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.58	18.57	464.16	2,662.82
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	245.23	25.15	289.25	1,127.69
Nutrient balance	56.35	-6.59	174.91	1,535.13
Applied to removed ratio	1.23	0.74	1.60	2.36

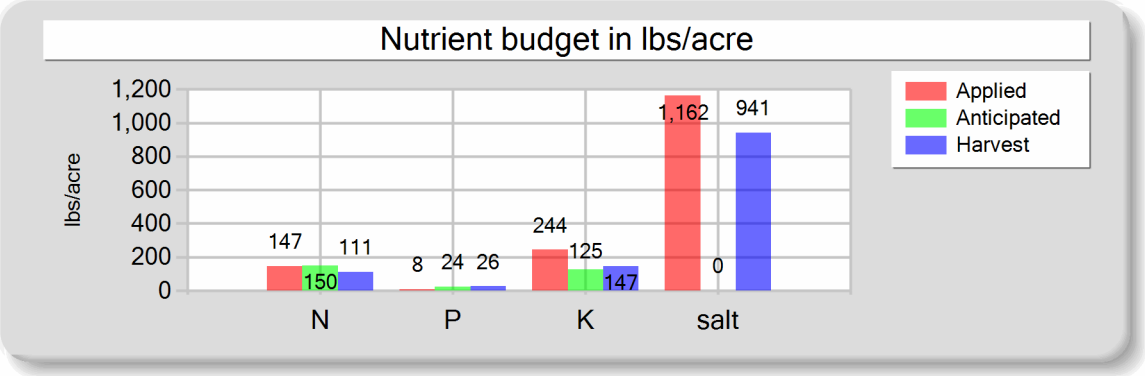
Fresh water applied	
16,926,000.00	*gallons*
623.33	*acre-inches*
31.17	*inches/acre*

Process wastewater applied	
1,386,000.00	*gallons*
51.04	*acre-inches*
2.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1-4 - 11/23/2022: Oats, silage-soft dough

Field name: D1-4 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	139.62	7.77	243.72	1,162.12
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.62	7.77	243.72	1,162.12
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	111.10	25.79	147.06	941.46
Nutrient balance	35.52	-18.03	96.65	220.67
Applied to removed ratio	1.32	0.30	1.66	1.23

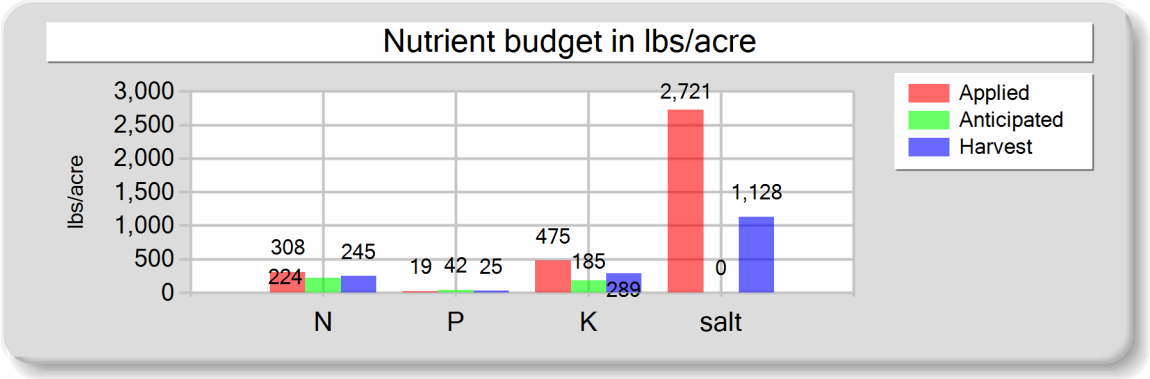
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
231,000.00	*gallons*
8.51	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1-4 - 06/15/2023: Corn, silage

Field name: D1-4 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	300.56	18.90	474.61	2,303.01
Fresh water	0.00	0.00	0.00	417.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.56	18.90	474.61	2,720.98
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	245.23	25.15	289.25	1,127.69
Nutrient balance	62.33	-6.25	185.36	1,593.29
Applied to removed ratio	1.25	0.75	1.64	2.41

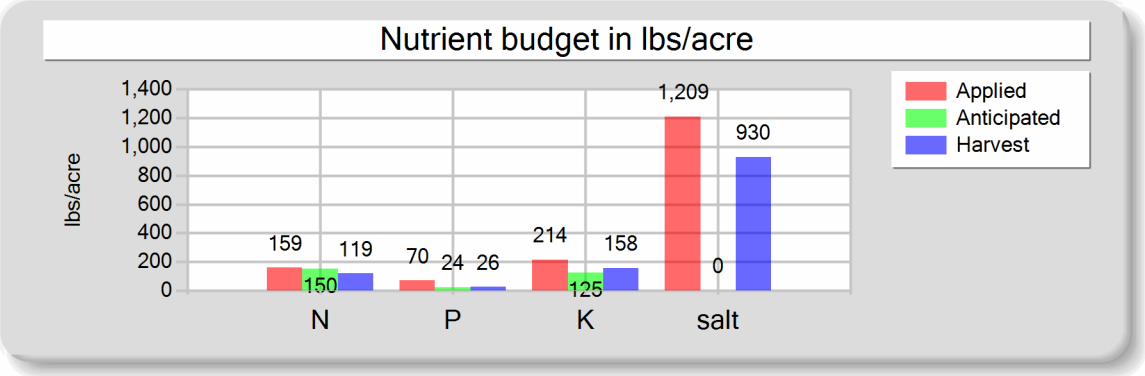
Fresh water applied	
6,045,000.00	*gallons*
222.62	*acre-inches*
31.80	*inches/acre*

Process wastewater applied	
495,000.00	*gallons*
18.23	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manteca Xavier - 11/21/2022: Oats, silage-soft dough

Field name: Manteca Xavier Crop: Oats, silage-soft dough Plant date: 11/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	151.62	70.18	213.60	1,208.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.62	70.18	213.60	1,208.88
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	119.23	25.63	158.11	929.70
Nutrient balance	39.39	44.54	55.49	279.18
Applied to removed ratio	1.33	2.74	1.35	1.30

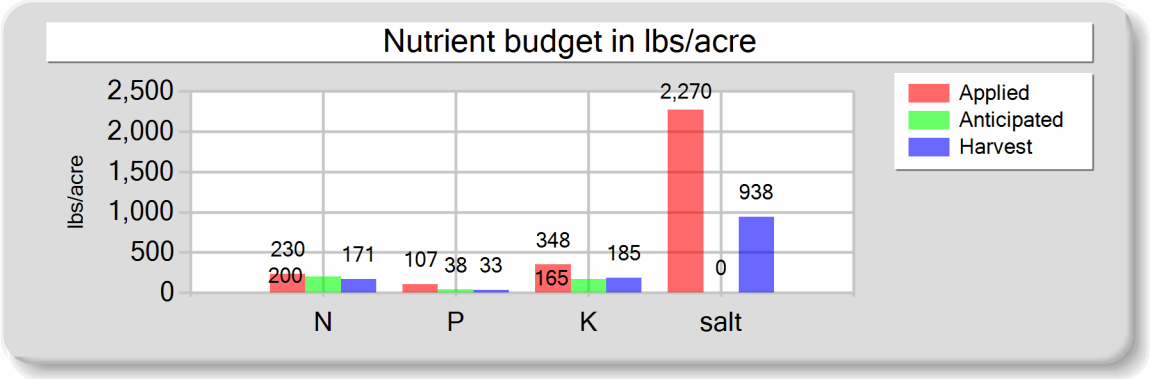
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Manteca Xavier - 06/20/2023: Corn, grain

Field name: Manteca Xavier Crop: Corn, grain Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	215.30	106.94	347.74	1,827.25
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	442.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.30	106.94	347.74	2,270.03
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	170.54	32.74	185.33	938.49
Nutrient balance	59.76	74.21	162.42	1,331.54
Applied to removed ratio	1.35	3.27	1.88	2.42

Fresh water applied	
77,760,000.00	*gallons*
2,863.64	*acre-inches*
33.69	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	13,700.00	41,700.00	25,600.00	14,000.00	12,200.00	11,700.00	2,160.00		23.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost

Sample and source description: Compost

Sample date: 04/17/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	9,300.00	10,600.00							54.70
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,400.00	15,100.00	49,100.00	25,100.00	13,800.00	17,900.00	7,200.00	3,830.00		25.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost

Sample and source description: Compost

Sample date: 10/09/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	5,930.00	6,760.00							71.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	9,640.00	29,500.00							23.60
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	365.00	0.00	0.00	574.00	667.00								6,090.00	5,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	496.00	314.00	0.00	0.00	92.80	595.00	185.00	122.00	170.00	2,700.00	0.00	368.00	307.00	5,880.00	5,040
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	507.00	283.00	0.00	0.00	28.20	885.00								5,450.00	4,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	512.00	367.00	0.00	0.00	36.40	712.00								6,460.00	3,540
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	476.00	300.00	0.00	0.00	50.80	626.00								5,940.00	4,080
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal SSJID

Canal SSJID

Sample description: Canal SSJID

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								89.20	58
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

D1-1

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

D1-2

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			121.00				
DL			4.00				

D1-3

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

D1-4

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

Manteca Xavier

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			79.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

D1-1 - 11/20/2022: Oats, silage-soft dough

D1-1 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,880.00	910.00	9,670.00		10.10
DL	100.00	100.00	100.00		0.01

D1-1 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	590.00	4,760.00		5.76
DL	100.00	100.00	100.00		0.01

D1-2 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	970.00	7,310.00		9.19
DL	100.00	100.00	100.00		0.01

D1-2 - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,530.00	560.00	3,220.00		3.93
DL	100.00	100.00	100.00		0.01

D1-3 - 11/23/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,480.00	1,040.00	5,930.00		8.55
DL	100.00	100.00	100.00		0.01

D1-3 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,680.00	480.00	5,520.00		5.88
DL	100.00	100.00	100.00		0.01

D1-4 - 11/23/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,480.00	1,040.00	5,930.00		8.55
DL	100.00	100.00	100.00		0.01

D1-4 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,680.00	480.00	5,520.00		5.88
DL	100.00	100.00	100.00		0.01

Manteca Xavier - 11/21/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,140.00	890.00	5,490.00		8.82
DL	100.00	100.00	100.00		0.01

Manteca Xavier - 06/20/2023: Corn, grain

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,230.00	620.00	3,510.00		5.29
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

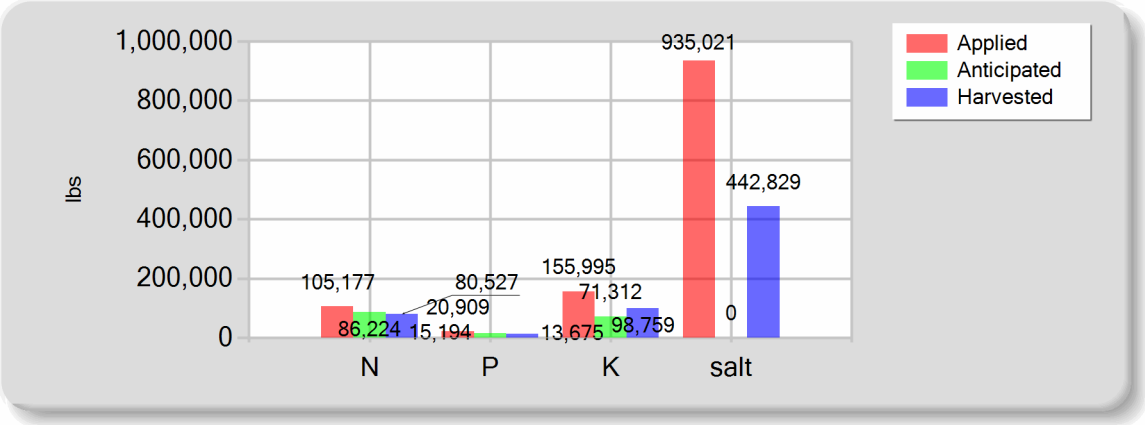
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

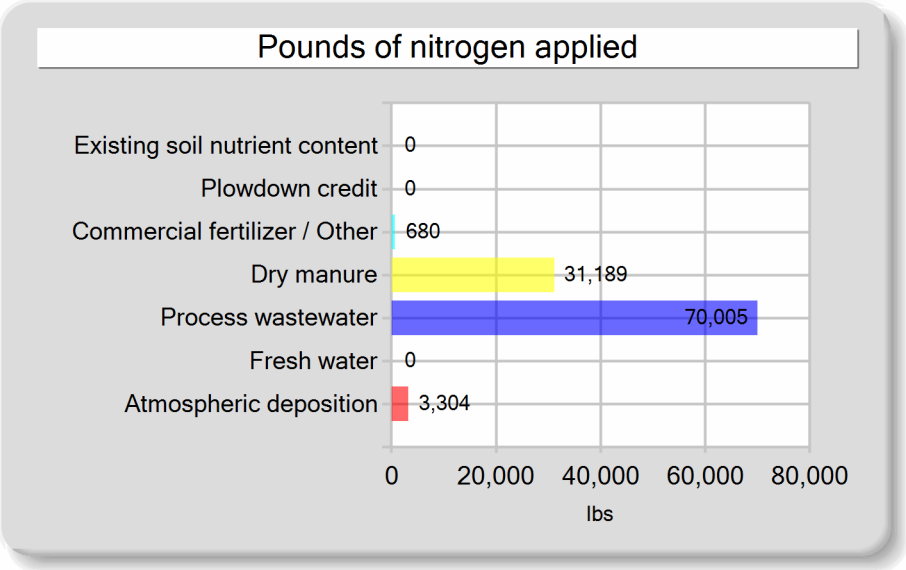
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

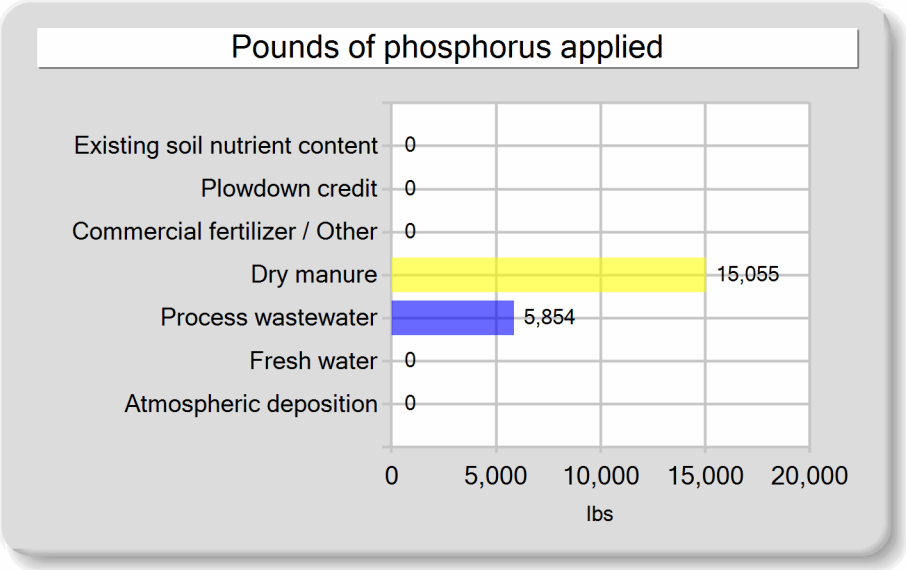
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	680.00	0.00	0.00	0.00
Dry manure	31,188.64	15,055.18	47,714.38	258,070.40
Process wastewater	70,004.79	5,853.53	108,280.56	577,757.73
Fresh water	0.00	0.00	0.00	99,192.53
Atmospheric deposition	3,304.00	0.00	0.00	0.00
Total nutrients applied	105,177.43	20,908.71	155,994.94	935,020.66
Anticipated crop nutrient removal	86,224.00	15,193.50	71,311.80	0.00
Actual crop nutrient removal	80,527.27	13,675.32	98,759.27	442,829.26
Nutrient balance	24,650.16	7,233.39	57,235.67	492,191.40
Applied to removed ratio	1.31	1.53	1.58	2.11

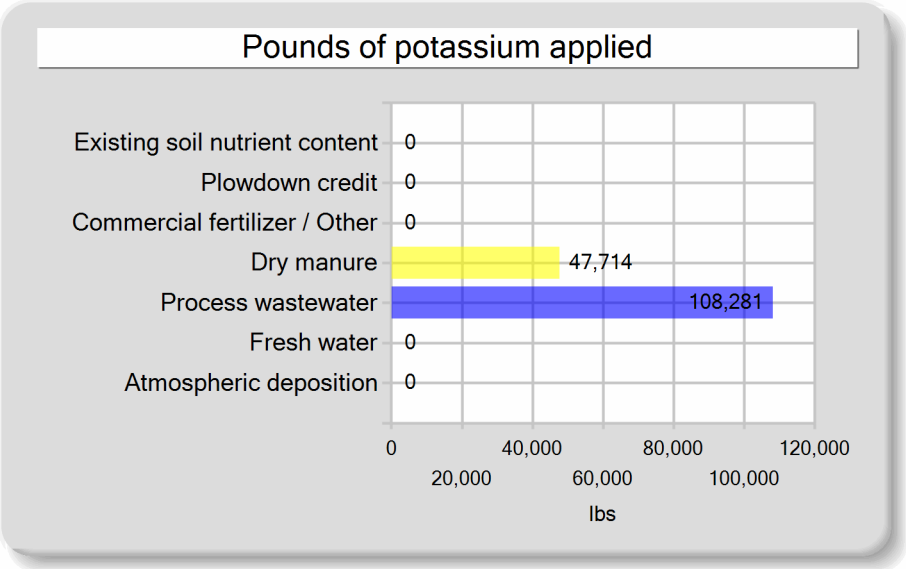
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

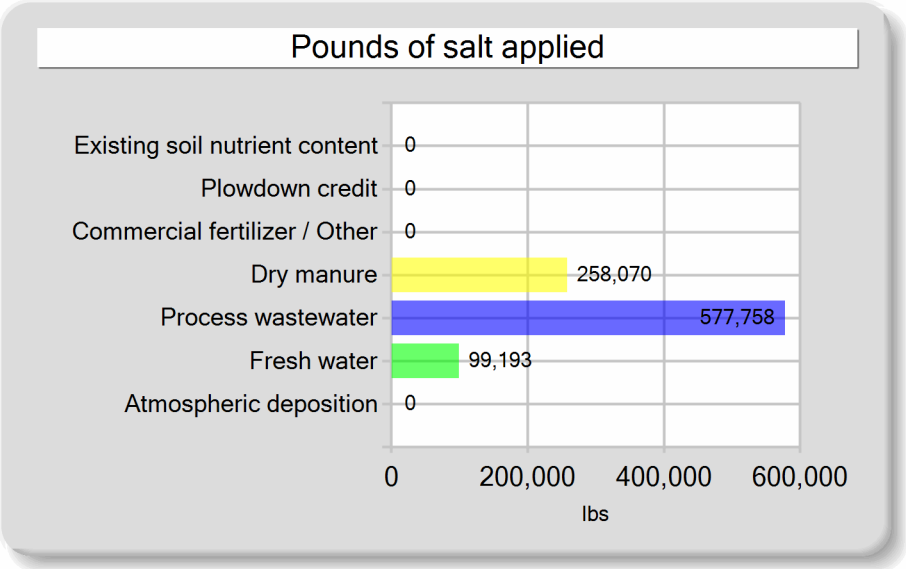


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mariana Brasil (Jun 19, 2024 08:58 PDT)	Dustin Wagner (Jun 19, 2024 15:07 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brasil & Sons Dairy	Dustin Wagner
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 19, 2024	Jun 19, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Dustin Wagner

Name of Dairy Facility: Wagner Dairy #2

Facility Address: 20623 Lone Tree Road, (Number and Street) Escalon, CA (City) 95320 (Zip Code)

Contact Person Name and Phone Number: Dustin Wagner (Name) 209-652-5192 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Brasil & Sons

Address of Hauling Company /Person: 21768 E. Lone Tree Road, Escalon, CA 95320 (Number and Street, City, Zip Code)

Contact Person: Victor Brasil (Name) 209-838-1661 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Brasil & Sons Dairy #2, 21728 Lone Tree Road, Escalon, CA 95320 209-595-1577

(Name, Number and Street, City, Zip Code, Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

21728 Lone Tree Road, (Number and Street) Escalon, CA (City) 95320 (Zip Code) 205-240-19 (Assessor's Parcel Number)

Dates Hauled: 5/23/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1070 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 57% moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

107 loads at 10 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Dustin Wagner (Dustin Wagner (Jun 19, 2024 15:07 PDT)) Date: Jun 19, 2024

Hauler's Signature: Victor Brasil Date: 6-10-24

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Wagner Dairy
22176 Skiff Rd.
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #2
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeanette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
North	T3349003A	Electrical Conductivity (EC)	EPA 120.1 and	0.520	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.84	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	333	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Turlock, CA 95382

PURCHASE ORDER: #2
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeanette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
South	T3349003B	Electrical Conductivity (EC)	EPA 120.1 and	0.592	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.10	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	379	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Wagner #2
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To:
F&R AG SERVICES

Sampled By: George
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
____ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/15/23		North	Domestic	10.4		✓				
2	↓		South	↓	10.2		✓				
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
	Jeanette Sawyer	F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	12-15-23 @ 11:30			T3349003
		(NO)				

(LABORATORY USE ONLY)

Wagner Dairy #2 2023 Annual Report

Final Audit Report 2024-06-19

Created:	2024-06-19
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAtP0nEZQBq-_8RjW1t2gd9wYqihPCtRTN

"Wagner Dairy #2 2023 Annual Report" History

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Signer wagnerdairylp@yahoo.com entered name at signing as Dustin Wagner
2024-06-19 - 10:07:31 PM GMT

Document e-signed by Dustin Wagner (wagnerdairylp@yahoo.com)
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